Red Team Internship Report

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Task 1: Passive Reconnaissance & Subdomain Enumeration

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Executive Summary

This report presents the findings from the passive reconnaissance phase conducted as part of the Red Team internship program. The primary objective of this exercise was to identify and map the publicly exposed assets associated with the target domain without directly interacting with its infrastructure. Multiple OSINT-based tools were used to perform subdomain enumeration and collect intelligence from publicly available sources. The results obtained from different tools were consolidated, deduplicated, and analyzed to build a clearer view of the target's external attack surface. This process helps reveal potentially overlooked or forgotten assets that could expand the organization's exposure. The identified subdomains provide a foundation for further infrastructure analysis and security assessment activities in later stages of the engagement.

Objective

The objective of this task was to conduct passive reconnaissance on the target domain to identify its publicly exposed assets. This was done by discovering subdomains using multiple OSINT-based enumeration tools without directly interacting with the target's infrastructure. The collected data helps in understanding the organization's external attack surface and identifying potential entry points. The findings from this phase provide a foundation for further analysis and security assessment in later stages.

Tools Used

The following tools were used to perform passive reconnaissance and subdomain enumeration during this task:

- **Subfinder** – Used to passively discover subdomains from multiple public data sources.
- **Assetfinder** – Helps identify subdomains associated with the target domain using OSINT sources.

- **OWASP Amass** – Performs in-depth passive enumeration and maps the target's external infrastructure.
- **DNSDumpster** – A web-based tool used to gather DNS records and discover additional subdomains.
- **crt.sh** – Queries certificate transparency logs to identify historical and hidden subdomains.

Methodology

Subdomain Discovery:

Subfinder:

subfinder -d hackthissite.org -silent -o subhack

```
┌──(kali㉿kali)-[~]
└─$ subfinder -d hackthissite.org -silent -o subhack

hp.hackthissite.org
irc-www.hackthissite.org
hub.irc.hackthissite.org
email.hackthissite.org
0-v3stage-cdn.hackthissit    g
irc-lille.hackthissite.org
irc-v6.hackthissite.org
mta-sts.hackthissite.org
h5ai.hackthissite.org
ns1.hackthissite.org
v3stage.hackthissite.org
ns2.hackthissite.org
irc.hackthissite.org
3-v3dev-cdn.hackthissite.org
vm-099.outbound.firewall.hackthissite.org
pi.hackthissite.org
psrp.hackthissite.org
vm-150.outbound.firewall.hackthissite.org
wolf.irc.hackthissite.org
www.irc.hackthissite.org
stats.hackthissite.org
lille.irc-v6.hackthissite.org
v3dev-cdn.hackthissite.or
lille.irc.hackthissite.org
mirror.hackthissite.org
status.hackthissite.org
vm-200.outbound.firewall.hackthissite.org
shadow.hackthissite.org
2-v3dev-cdn.hackthissite.org
4-v3dev-cdn.hackthissite.org
4-v3stage-cdn.hackthissite.org
daemon.hackthissite.org
irc-wolf.hackthissite.org
new-irc.hackthissite.org
irc-hub.hackthissite.org
vm-050.outbound.firew  ll.hackthissite.org
api.hackthissite.org
git.hackthissite.org
www.hackthissite.org
0-v3dev-cdn.hackthissite.org
```

AssetFinder :

assetfinder --subs-only hackthissite.org > assethack.txt

```
┌──(kali㉿kali)-[~/Week 1]
└─$ assetfinder --subs-only hackthissite.org > assethack.txt
```

cat assethack.txt

```
┌──(kali㉿kali)-[~/Week1]
└─$ cat assethack.txt
hackthissite.com
www.hackthissite.com
hackthissite.com
www.hackthissite.com
```

Amass :

amass enum -passive -d hackthissite.org -o amass.txt

```
┌──(kali㉿kali)-[~/Week 1]
└─$ amass enum -passive -d hackthissite.org -o amass.txt
hackthissite.org (FQDN) --> mx_record --> aspmx.l.google.com (FQDN)
hackthissite.org (FQDN) --> mx_record --> aspmx5.googlemail.com (FQDN)
hackthissite.org (FQDN) --> mx_record --> aspmx4.googlemail.com (FQDN)
hackthissite.org (FQDN) --> mx_record --> alt1.aspmx.l.google.com (FQDN)
hackthissite.org (FQDN) --> mx_record --> aspmx3.googlemail.com (FQDN)
hackthissite.org (FQDN) --> mx_record --> alt2.aspmx.l.google.com (FQDN)
hackthissite.org (FQDN) --> mx_record --> aspmx2.googlemail.com (FQDN)
hackthissite.org (FQDN) --> ns_record --> f.ns.buddyns.com (FQDN)
hackthissite.org (FQDN) --> ns_record --> j.ns.buddyns.com (FQDN)
hackthissite.org (FQDN) --> ns_record --> c.ns.buddyns.com (FQDN)
hackthissite.org (FQDN) --> ns_record --> h.ns.buddyns.com (FQDN)
hackthissite.org (FQDN) --> ns_record --> g.ns.buddyns.com (FQDN)
hackthissite.org (FQDN) --> node --> status.hackthissite.org (FQDN)
hackthissite.org (FQDN) --> node --> 3-v3stage-cdn.hackthissite.org (FQDN)
hackthissite.org (FQDN) --> node --> v3stage-cdn.hackthissite.org (FQDN)
hackthissite.org (FQDN) --> node --> forum.hackthissite.org (FQDN)
hackthissite.org (FQDN) --> node --> v3dev.hackthissite.org (FQDN)
hackthissite.org (FQDN) --> node --> v3dev-cdn.hackthissite.org (FQDN)
hackthissite.org (FQDN) --> node --> qdb.hackthissite.org (FQDN)
hackthissite.org (FQDN) --> node --> irc-www.hackthissite.org (FQDN)
hackthissite.org (FQDN) --> node --> 2-v3stage-cdn.hackthissite.org (FQDN)
hackthissite.org (FQDN) --> node --> psrp.hackthissite.org (FQDN)
status.hackthissite.org (FQDN) --> cname_record --> edge.redirect.pizza (FQDN)
3-v3stage-cdn.hackthissite.org (FQDN) --> cname_record --> v3stage-cdn.hackthissite.org (FQDN)
forum.hackthissite.org (FQDN) --> cname_record --> hackthissite.org (FQDN)
v3dev-cdn.hackthissite.org (FQDN) --> cname_record --> v3dev.hackthissite.org (FQDN)
qdb.hackthissite.org (FQDN) --> cname_record --> irc-www.hackthissite.org (FQDN)
2-v3stage-cdn.hackthissite.org (FQDN) --> cname_record --> v3stage-cdn.hackthissite.org (FQDN)
psrp.hackthissite.org (FQDN) --> cname_record --> hackmail.org (FQDN)
hackthissite.org (FQDN) --> node --> 0-v3dev-cdn.hackthissite.org (FQDN)
hackthissite.org (FQDN) --> node --> www.irc.hackthissite.org (FQDN)
hackthissite.org (FQDN) --> node --> h5ai.hackthissite.org (FQDN)
0-v3dev-cdn.hackthissite.org (FQDN) --> cname_record --> v3dev.hackthissite.org (FQDN)
irc-www.hackthissite.org (FQDN) --> cname_record --> www.irc.hackthissite.org (FQDN)
```

Data Merging :

Now by Merging all the data of subdomains that we collect by a simple command i.e:

cat subhack.txt assethack.txt amass.txt | sort -u > all_subdomains.txt

```
┌──(kali㉿kali)-[~/Week 1]
└─$ cat amass.txt assethack.txt subhack.txt | sort -u > all_subdomains.txt
```

- **cat** combines the contents of multiple files into one stream.
- **sort -u** sorts the entries alphabetically and removes duplicate records.
- The final output is saved in **all_subdomains.txt**, which represents the consolidated list of unique subdomains.

Asset Counting:

Command used:

wc -l all_subdomains.txt

```
┌──(kali㉿kali)-[~/Week 1]
└─$ wc -l all_subdomains.txt
360 all_subdomains.txt
```

Confidence Analysis:

To evaluate the reliability of discovered subdomains, the overlap between results returned by different tools was analyzed. Subdomains appearing in multiple sources are generally more likely to be valid and active assets.

Command used:

sort assethack.txt amass.txt subhack.txt | uniq -c | sort -nr | head

```
┌──(kali㉿kali)-[~/Week 1]
└─$ sort assethack.txt amass.txt subhack.txt| uniq -c | sort -nr | head
      2 www.hackthissite.com
      2 hackthissite.com
      1 www.irc.hackthissite.org
      1 www.hackthissite.org
      1 wolf.irc-v6.hackthissite.org
      1 wolf.irc.hackthissite.org
      1 vm-200.outbound.firewall.hackthissite.org
      1 vm-150.outbound.firewall.hackthissite.org
      1 vm-099.outbound.firewall.hackthissite.org
      1 vm-050.outbound.firewall.hackthissite.org
```

- **sort** organizes the combined results.
- **uniq -c** counts how many times each subdomain appears across the files.

- **sort -nr** sorts the counts in descending order.
- **head** displays the most frequently repeated subdomains.

Historical Subdomain Discovery:

Crt.sh:

This platform queries **certificate transparency logs** to identify domains that have been issued SSL/TLS certificates.

Command Used:

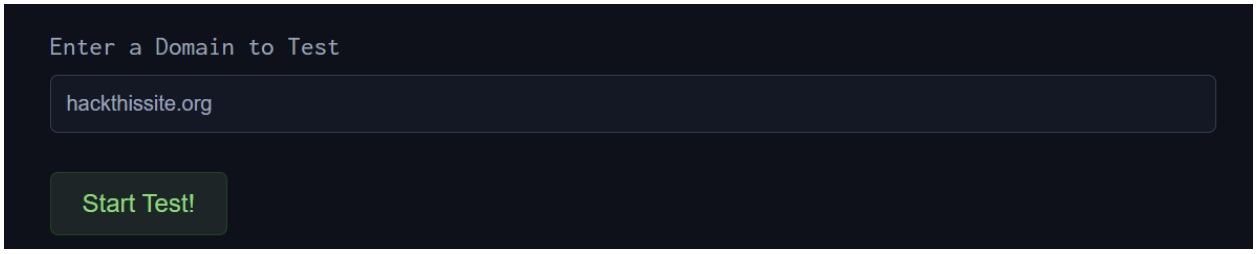
Crt_v2.sh -d hackthissite.org > crt_hack.txt

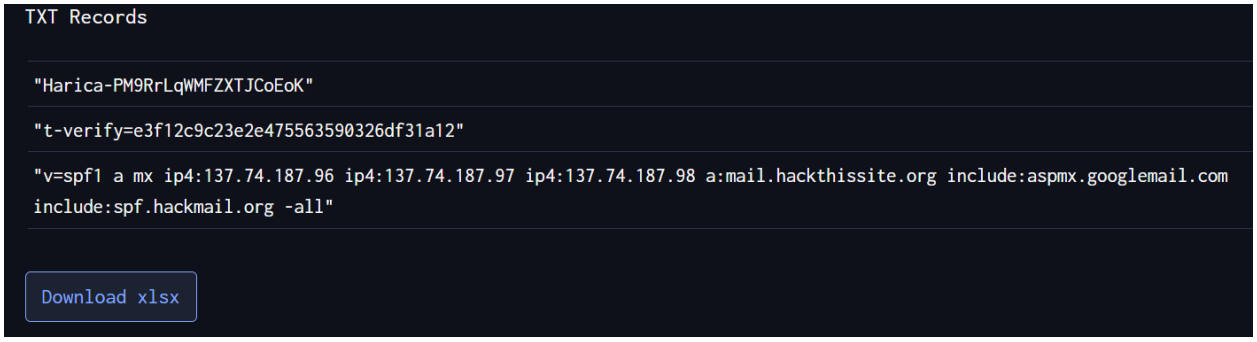
```
┌──(kali㉿kali)-[~/Week 1]
└─$ crt_v2.sh -d hackthissite.org > crt_hack.txt

┌──(kali㉿kali)-[~/Week 1]
└─$ cat crt_hack.txt

+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|      ..| search crt.sh v 2.0 |..     |
+   site : crt.sh Certificate Search  +
|          Twitter: az7rb            |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+

ctf.hackthissite.org
email.hackthissite.org
h5ai.hackthissite.org
hacked.chat
hackmail.org
hacksite.net
hackthisjogneh42n5o7gbzrewxee3vyu6ex37ukyvdw6jm66npakiyd.onion
hackthissite.org
irc.hackthissite.org
irc-ipv6.hackthissite.org
irc-v6.hackthissite.org
irc-wolf.hackthissite.org
lille.irc.hackthissite.org
lille.irc-v6.hackthissite.org
mail.hackthissite.org
mail.infosecedu.org
mta-sts.hackthissite.org
new-irc.hackthissite.org
psrp.hackthissite.org
status.hackthissite.org
status-new.hackthissite.org
wolf.irc.hackthissite.org
wolf.irc-v6.hackthissite.org
www.hackthissite.org
www.irc.hackthissite.org

[+] Total Save will be 25 Domain only
```

DNSDumpster:

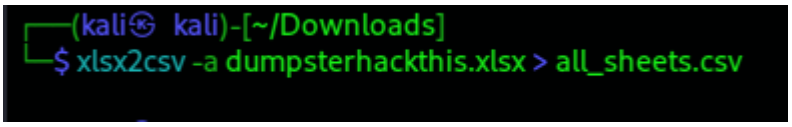
DNSDumpster is an online reconnaissance tool that collects DNS records related to a domain. It can reveal additional subdomains, IP addresses, and infrastructure details that may not appear in standard enumeration results.

- Go to the website, search for the target domain, start the test, and download the data in .xlsx format.





- Now download the xlsx2csv by running command “apt install xlsx2csv”
- Convert the xlsx to csv by command i.e “xlsx2csv -a dumpsterhackthis.xlsx > all_sheets.csv



- After converting the file, pattern matching was used to extract subdomains related to the target domain and store them in a separate file by using this command,

  grep -oE "[a-zA-Z0-9.-]+\.hackthissite\.org" all_sheets.csv | sort -u > dnsdump_hackthis.txt

  cat dnsdump_hackthis.txt

```
┌──(kali㉿kali)-[~/Downloads]
└─$ grep -oE "[a-zA-Z0-9.-]+\.hackthissite\.org" all_sheets.csv | sort -u > dnsdump_hackt

┌──(kali㉿kali)-[~/Downloads]
└─$ cat dnsdump_hackthis.txt
api.hackthissite.org
ctf.hackthissite.org
email.hackthissite.org
forums.hackthissite.org
git.hackthissite.org
h5ai.hackthissite.org
hp.hackthissite.org
irc.hackthissite.org
irc-hub.hackthissite.org
irc-www.hackthissite.org
legal.hackthissite.org
lille.irc.hackthissite.org
mirror.hackthissite.org
psrp.hackthissite.org
qdb.hackthissite.org
stats.hackthissite.org
status.hackthissite.org
wolf.irc.hackthissite.org
www.hackthissite.org
www.irc.hackthissite.org
```

- The results obtained from **DNSDumpster** were then merged with subdomain lists generated by other enumeration tools to create a final consolidated dataset.

  cat all_subdomains.txt crt_hack.txt dnsdump_hackthis.txt | sort –u > allattacks.txt

  cat allattacks.txt

```
┌──(kali㉿kali)-[~/Week 1]
└─$ cat all_subdomains.txt dnsdump_hackthis.txt crt_hack.txt| sort -u > allattacks.txt

┌──(kali㉿kali)-[~/Week 1]
└─$ cat allattacks.txt

+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
0-v3dev-cdn.hackthissite.org
0-v3stage-cdn.hackthissite.org
1-v3dev-cdn.hackthissite.org
1-v3stage-cdn.hackthissite.org
2-v3dev-cdn.hackthissite.org
2-v3stage-cdn.hackthissite.org
[+] Output saved in output/domain.hackthissite.org.txt
[+] Total Save will be 25 Domain only
3-v3dev-cdn.hackthissite.org
3-v3stage-cdn.hackthissite.org
4-v3dev-cdn.hackthissite.org
4-v3stage-cdn.hackthissite.org
0.0.1.0.7.8.1.0.4.7.0.0.7.3.1.0.8.d.c.c.8.0.0.0.0.d.1.4.1.0.0.2.ip6.arpa (FQDN) --> ptr_record --> hackthissite.org (FQDN)
0 (ASN) --> announces --> 2406:d500:7::/48 (Netblock)
0 (ASN) --> announces --> 5.220.0.0/14 (Netblock)
0 (ASN) --> managed_by --> Not routed (RIROrganization)
0-v3dev-cdn.hackthissite.org (FQDN) --> cname_record --> v3dev.hackthissite.org (FQDN)
0-v3stage-cdn.hackthissite.org (FQDN) --> cname_record --> v3stage-cdn.hackthissite.org (FQDN)
100.187.74.137.in-addr.arpa (FQDN) --> ptr_record --> hackthissite.org (FQDN)
1.0.1.0.7.8.1.0.4.7.0.0.7.3.1.0.8.d.c.c.8.0.0.0.0.d.1.4.1.0.0.2.ip6.arpa (FQDN) --> ptr_record --> hackthissite.org (FQDN)
101.187.74.137.in-addr.arpa (FQDN) --> ptr_record --> hackthissite.org (FQDN)
102.187.74.137.in-addr.arpa (FQDN) --> ptr_record --> hackthissite.org (FQDN)
103.187.74.137.in-addr.arpa (FQDN) --> ptr_record --> hackthissite.org (FQDN)
103.25.56.0/22 (Netblock) --> contains --> 103.25.56.55 (IPAddress)
104.187.74.137.in-addr.arpa (FQDN) --> ptr_record --> hackthissite.org (FQDN)
108.177.0.0/17 (Netblock) --> contains --> 108.177.125.27 (IPAddress)
115.187.74.137.in-addr.arpa (FQDN) --> ptr_record --> api.hackthissite.org (FQDN)
116.187.74.137.in-addr.arpa (FQDN) --> ptr_record --> api.hackthissite.org (FQDN)
116.202.0.0/15 (Netblock) --> contains --> 116.203.6.3 (IPAddress)
117.187.74.137.in-addr.arpa (FQDN) --> ptr_record --> api.hackthissite.org (FQDN)
118.187.74.137.in-addr.arpa (FQDN) --> ptr_record --> api.hackthissite.org (FQDN)
119.187.74.137.in-addr.arpa (FQDN) --> ptr_record --> api.hackthissite.org (FQDN)
123.187.74.137.in-addr.arpa (FQDN) --> ptr_record --> pi.hackthissite.org (FQDN)
133.187.74.137.in-addr.arpa (FQDN) --> ptr_record --> mirror.hackthissite.org (FQDN)
```

Conclusion:

The passive reconnaissance phase successfully identified multiple subdomains related to the target domain using several OSINT-based enumeration techniques. By combining results from different tools and removing duplicates, a consolidated list of unique assets was created to represent the organization's external attack surface. Historical data sources further helped expand the dataset and reveal additional infrastructure details.

The findings from this phase provide a valuable foundation for future stages of security assessment, where the discovered assets can be analyzed further for potential vulnerabilities or misconfigurations.

Task 02 – Infrastructure & Technology Profiling

Objective:

The objective of this task is to identify the technologies, infrastructure components, and hosting details associated with the target domain. By analyzing publicly available information such as web technologies, DNS records, and domain registration data, it becomes possible to understand how the target's systems are structured. This information helps security professionals map the environment and identify areas that may require further security assessment.

Tools Used:

- **WhatWeb** – Identifies technologies used by a website such as web servers, CMS platforms, and frameworks.
- **Wappalyzer** – Detects the technology stack used by web applications.
- **WHOIS** – Retrieves domain registration and ownership information.
- **nslookup** – Queries DNS records to identify IP addresses and DNS servers.
- **dig** – Provides detailed DNS record information.

Technology Fingerprinting:

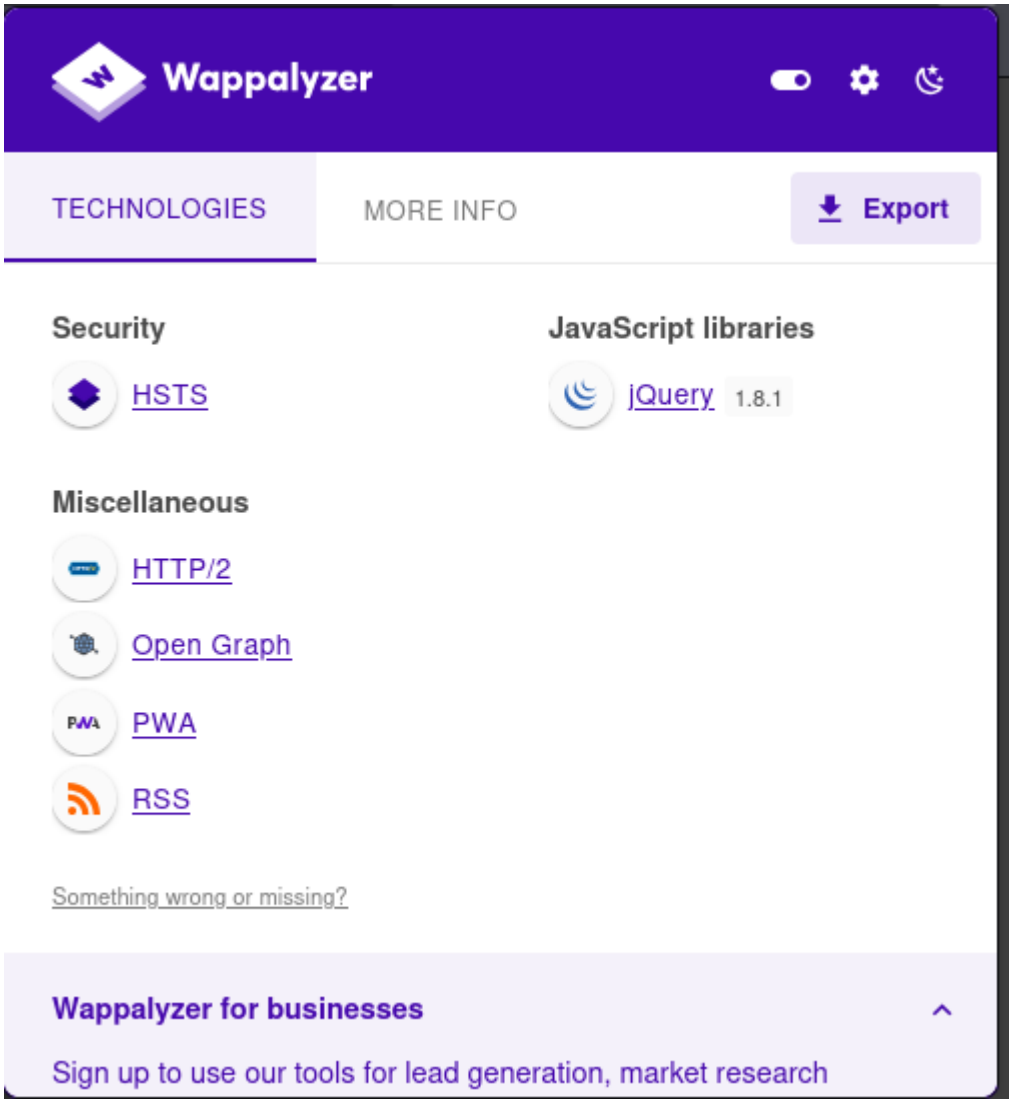
Using WhatWeb:

whatweb https://hackthissite.org

```
┌──(kali㉿kali)-[~]
└─$ whatweb https://hackthissite.org
https://hackthissite.org [200 OK] Content-Language[en], Cookies[HackThisSite], Country[CANADA][CA], HTTPServer[HackThisSite], IP[137.74.187.103], JQuery[1.8.1], Meta-Author[HackThisSite.org Staff], Open-Graph-P
rotocol[website], PasswordField[password], Script[text/javascript], Strict-Transport-Security[max-age=31536000; includeSubDomains; preload], Title[Hack This Site], UncommonHeaders[upgrade,onion-location,access-
control-allow-origin,content-security-policy,referrer-policy,feature-policy,public-key-pins-report-only,report-to,nel], X-XSS-Protection[0]
```

Using Wappalyzer:

To validate the results obtained from WhatWeb, **Wappalyzer** was used to identify additional technologies associated with the target website.



Wappalyzer analyzes several indicators such as HTTP headers, JavaScript files, and metadata to determine the technology stack used by the web application.

Domain Information Analysis:

Using WHOIS:

- whois hackthissite.org > whoishack.txt

```
┌──(kali㉿kali)-[~/Week 1]
└─$ whois hackthissite.org > whoishack.txt

┌──(kali㉿kali)-[~/Week 1]
└─$ cat whoishack.txt
Domain Name: hackthissite.org
Registry Domain ID: REDACTED
Registrar WHOIS Server: http://whois.porkbun.com
Registrar URL: https://porkbun.com
Updated Date: 2025-07-15T23:08:30Z
Creation Date: 2003-08-10T15:01:25Z
Registry Expiry Date: 2026-08-10T15:01:25Z
Registrar: Porkbun LLC
Registrar IANA ID: 1861
Registrar Abuse Contact Email: abuse@porkbun.com
Registrar Abuse Contact Phone: +1.8557675286
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Name Server: c.ns.buddyns.com
Name Server: f.ns.buddyns.com
Name Server: g.ns.buddyns.com
Name Server: h.ns.buddyns.com
Name Server: j.ns.buddyns.com
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://icann.org/wicf/
>>> Last update of WHOIS database: 2026-03-06T17:54:02Z <<<
```

The WHOIS query provides information such as:

- Domain registrar
- Organization name
- Name servers
- Registration dates
- Abuse contact information

DNS Record Analysis:

Using nslookup:

nslookup hackthissite.org

```
┌──(kali㉿kali)-[~/Week 1]
└─$ nslookup hackthissite.org
Server:         192.168.0.1
Address:        192.168.0.1#53

Name:   hackthissite.org
Address: 137.74.187.103
```

This command retrieves the IP address associated with the domain and helps identify the hosting location.

Additional DNS record queries can also be performed:

1. nslookup -type=MX hackthissite.org
2. nslookup -type=NS hackthissite.org
3. nslookup -type=TXT hackthissite.org

```
┌──(kali㉿kali)-[~/Week 1]
└─$ nslookup -type=MX hackthissite.org
Server:         2402:e000:422:ba80::39
Address:        2402:e000:422:ba80::39#53

Non-authoritative answer:
hackthissite.org        mail exchanger = 30 aspmx3.googlemail.com.
hackthissite.org        mail exchanger = 30 aspmx4.googlemail.com.
hackthissite.org        mail exchanger = 30 aspmx2.googlemail.com.
hackthissite.org        mail exchanger = 10 aspmx.l.google.com.
hackthissite.org        mail exchanger = 30 aspmx5.googlemail.com.
hackthissite.org        mail exchanger = 20 alt1.aspmx.l.google.com.
hackthissite.org        mail exchanger = 20 alt2.aspmx.l.google.com.

Authoritative answers can be found from:


┌──(kali㉿kali)-[~/Week 1]
└─$ nslookup -type=NS hackthissite.org
Server:         2402:e000:422:ba80::39
Address:        2402:e000:422:ba80::39#53

Non-authoritative answer:
hackthissite.org        nameserver = c.ns.buddyns.com.
hackthissite.org        nameserver = h.ns.buddyns.com.
hackthissite.org        nameserver = g.ns.buddyns.com.
hackthissite.org        nameserver = j.ns.buddyns.com.
hackthissite.org        nameserver = f.ns.buddyns.com.

Authoritative answers can be found from:
h.ns.buddyns.com        internet address = 103.25.56.55
c.ns.buddyns.com        internet address = 116.203.6.3
j.ns.buddyns.com        internet address = 37.143.61.179
g.ns.buddyns.com        internet address = 192.184.93.99
f.ns.buddyns.com        internet address = 5.223.55.119
h.ns.buddyns.com        has AAAA address 2406:d500:2::de4f:f105
c.ns.buddyns.com        has AAAA address 2a01:4f8:1c0c:8115::3
j.ns.buddyns.com        has AAAA address 2a01:a500:2766::5c3f:d10b
g.ns.buddyns.com        has AAAA address 2604:180:1:92a::3
f.ns.buddyns.com        has AAAA address 2a01:4ff:2f0:3661::3


┌──(kali㉿kali)-[~/Week 1]
└─$ nslookup -type=TXT hackthissite.org
Server:         2402:e000:422:ba80::39
Address:        2402:e000:422:ba80::39#53

Non-authoritative answer:
hackthissite.org        text = "t-verify=e3f12c9c23e2e475563590326df31a12"
hackthissite.org        text = "Harica-PM9RrLqWMFZXTJCoEoK"
hackthissite.org        text = "v=spf1 a mx ip4:137.74.187.96 ip4:137.74.187.97 ip4:137.74.187.98 a:mail.hackthissite.org include:aspmx.googlemail.com include:spf.hackmail.org -all"
```

Using dig:

The **dig** command provides more detailed DNS information.

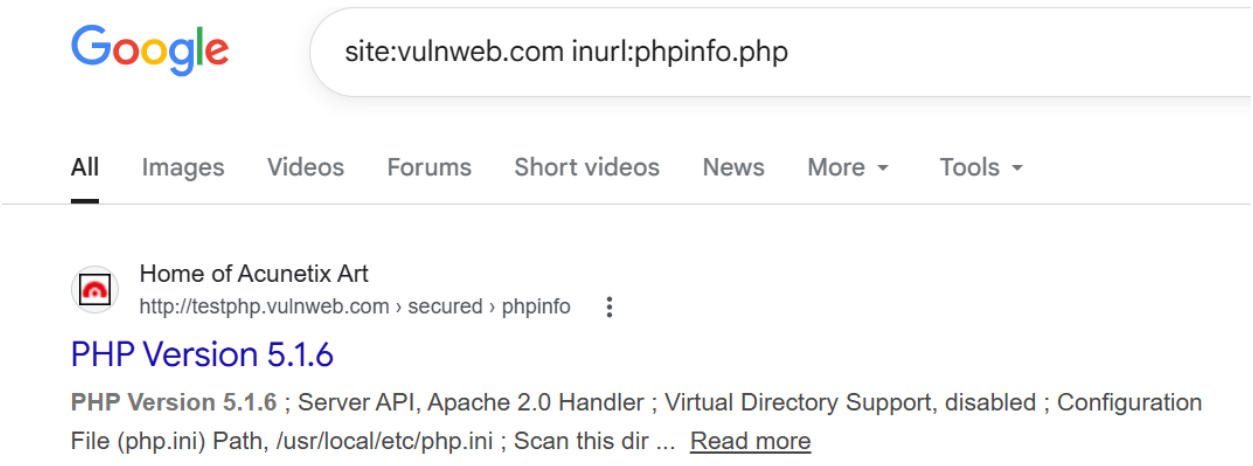
dig hackthissite.org ANY

```
┌──(kali㉿kali)-[~/Week 1]
└─$ dig hackthissite.org ANY

; <<>> DiG 9.20.18-1-Debian <<>> hackthissite.org ANY
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 23566
;; flags: qr rd ra; QUERY: 1, ANSWER: 29, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 867d9dce5583c54c0100000069ac6322d83db839a634b25f (good)
;; QUESTION SECTION:
;hackthissite.org.              IN      ANY

;; ANSWER SECTION:
hackthissite.org.       3600    IN      SOA     c.ns.buddyns.com. admin.hackthissite.org. 2025032501 3600 900 1209600 3600
hackthissite.org.       3600    IN      MX      20 alt1.aspmx.l.google.com.
hackthissite.org.       3600    IN      MX      20 alt2.aspmx.l.google.com.
hackthissite.org.       3600    IN      MX      30 aspmx2.googlemail.com.
hackthissite.org.       3600    IN      MX      30 aspmx4.googlemail.com.
hackthissite.org.       3600    IN      MX      30 aspmx3.googlemail.com.
hackthissite.org.       3600    IN      MX      10 aspmx.l.google.com.
hackthissite.org.       3600    IN      MX      30 aspmx5.googlemail.com.
hackthissite.org.       3600    IN      TXT     "v=spf1 a mx ip4:137.74.187.96 ip4:137.74.187.97 ip4:137.74.187.98 a:mail.hackthissite.org
hackthissite.org.       3600    IN      TXT     "Harica-PM9RrLqWMFZXTJCoEoK"
hackthissite.org.       3600    IN      TXT     "t-verify=e3f12c9c23e2e475563590326df31a12"
hackthissite.org.       3600    IN      A       137.74.187.101
hackthissite.org.       3600    IN      A       137.74.187.100
hackthissite.org.       3600    IN      A       137.74.187.103
hackthissite.org.       3600    IN      A       137.74.187.102
hackthissite.org.       3600    IN      A       137.74.187.104
hackthissite.org.       170     IN      NS      j.ns.buddyns.com.
```

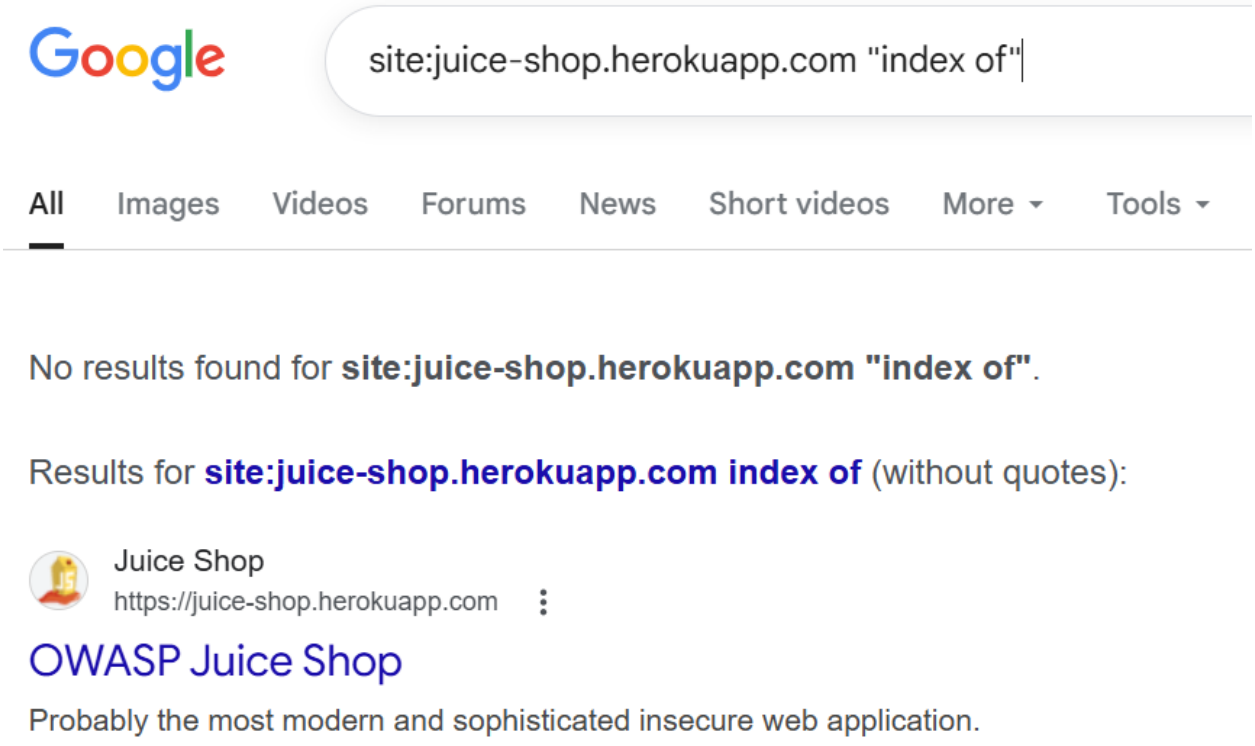
Advanced Google Dorking:

The following examples demonstrate how advanced search operators can be used to identify potentially sensitive files if they are publicly exposed.

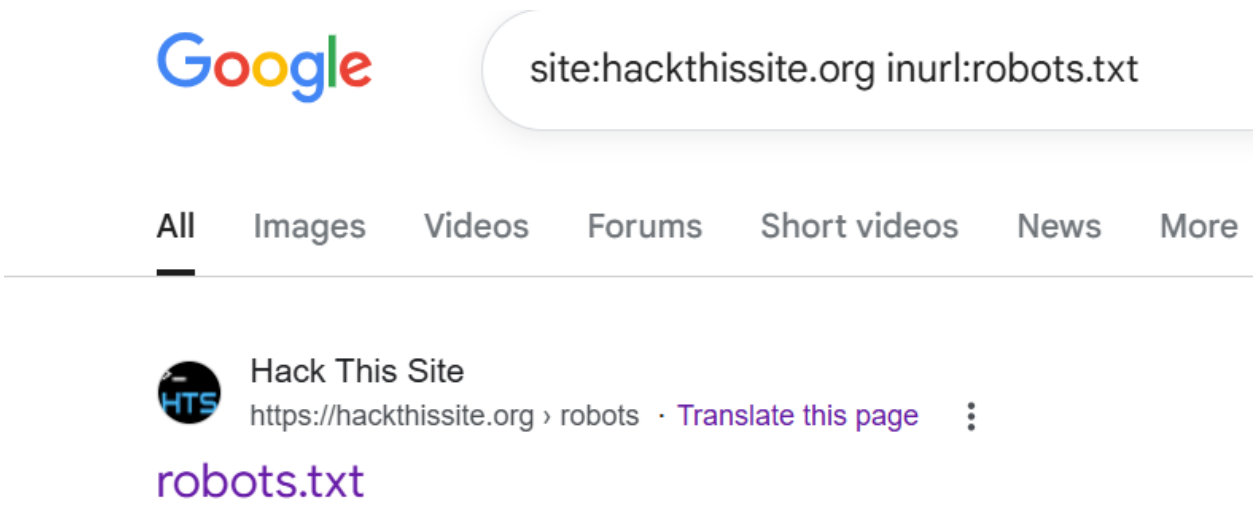
- site:vulnweb.com inurl:phpinfo.php



- site:juice-shop.herokuapp.com "index of"



- site:hackthissite.org inurl:robots.txt



Finding Backup Files (.bak, .zip, .old, .tar)

site:target.com filetype:bak
site:target.com filetype:zip
site:target.com inurl:backup
site:target.com ext:sql
site:target.com ext:old

Conclusion:

The infrastructure and technology profiling phase successfully identified several key components related to the target domain. By analyzing web technologies, DNS records, and domain registration data, a clearer understanding of the target's hosting environment and application stack was obtained. This information is valuable for identifying potential attack surfaces and preparing for deeper security assessments in later phases of the engagement.

Task 3 – GitHub Dorking & Credential Leaks

Objective:

The objective of this task is to identify sensitive information or credentials that may have been accidentally exposed in public repositories on GitHub. The task focuses on detecting hardcoded secrets such as API keys, passwords, and private keys using different search queries. The goal is to understand how exposed credentials can become a security risk if not properly managed.

Tools Used:

GitHub – Used to search public repositories for leaked credentials

WEB Browser – Used to perform manual GitHub searches

Methodology:

The methodology for this task involved searching public repositories on GitHub using specific keywords and queries to identify exposed credentials. Different search techniques were used, including organization-based searches and public searches related to the target domain.

Findings:

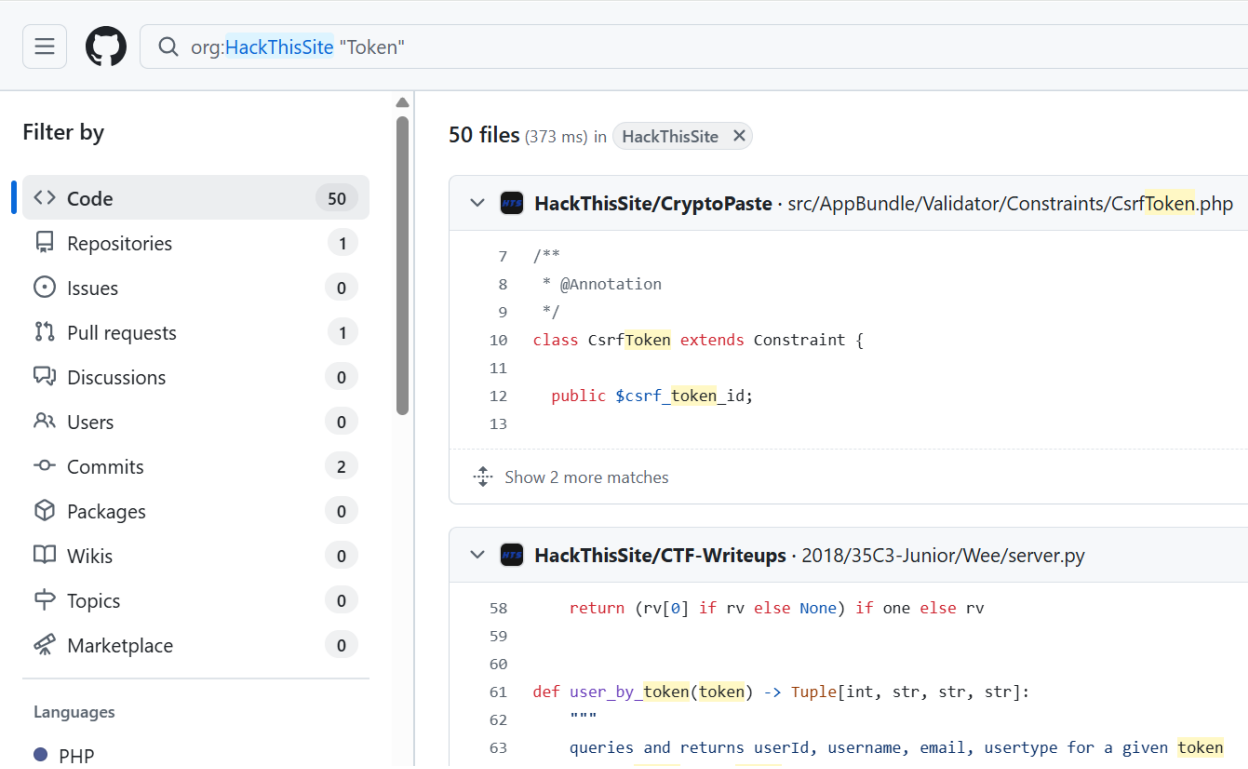
Searching for Tokens in Repositories:

A search was performed on GitHub using the following query:

org:HackThisSite "Token"

This search revealed multiple files within repositories belonging to the HackThisSite organization containing references to tokens. The results included files where variables such as csrf_token were defined in the source code. These

tokens are used for security mechanisms such as CSRF protection. Although they are not actual leaked credentials, they demonstrate how developers store security-related tokens within application code.

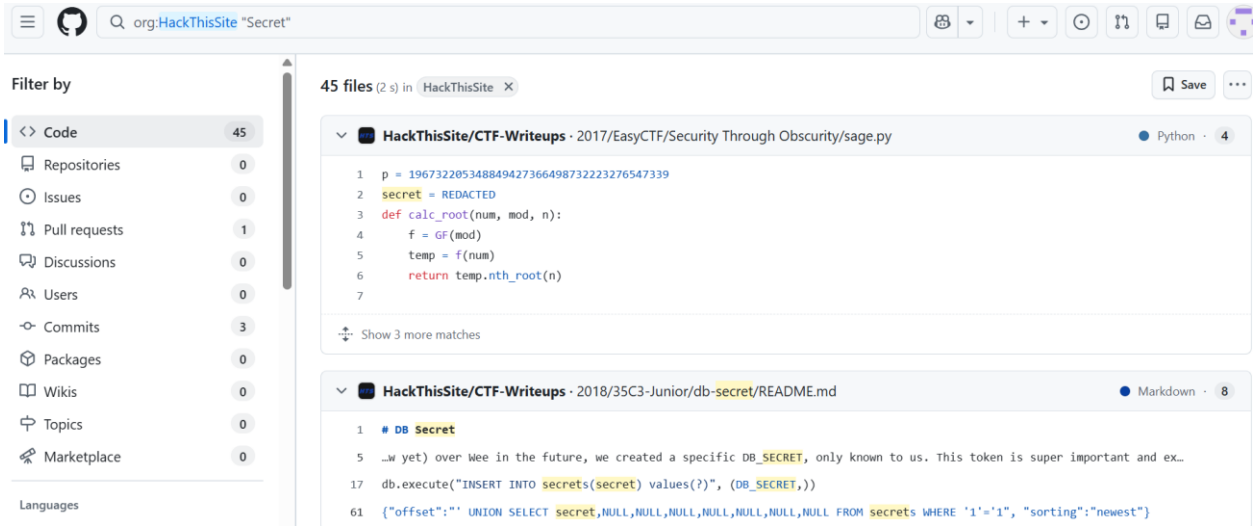


Searching for Secret Values:

Another search query was used to identify references to secrets:

org:HackThisSite "Secret"

The search returned several files referencing secret values within the repository. In some cases, the results included configuration or challenge files where secret variables were defined. While these values were part of training or challenge environments, the results demonstrate how searching for the keyword **"secret"** can help identify sensitive information in repositories.

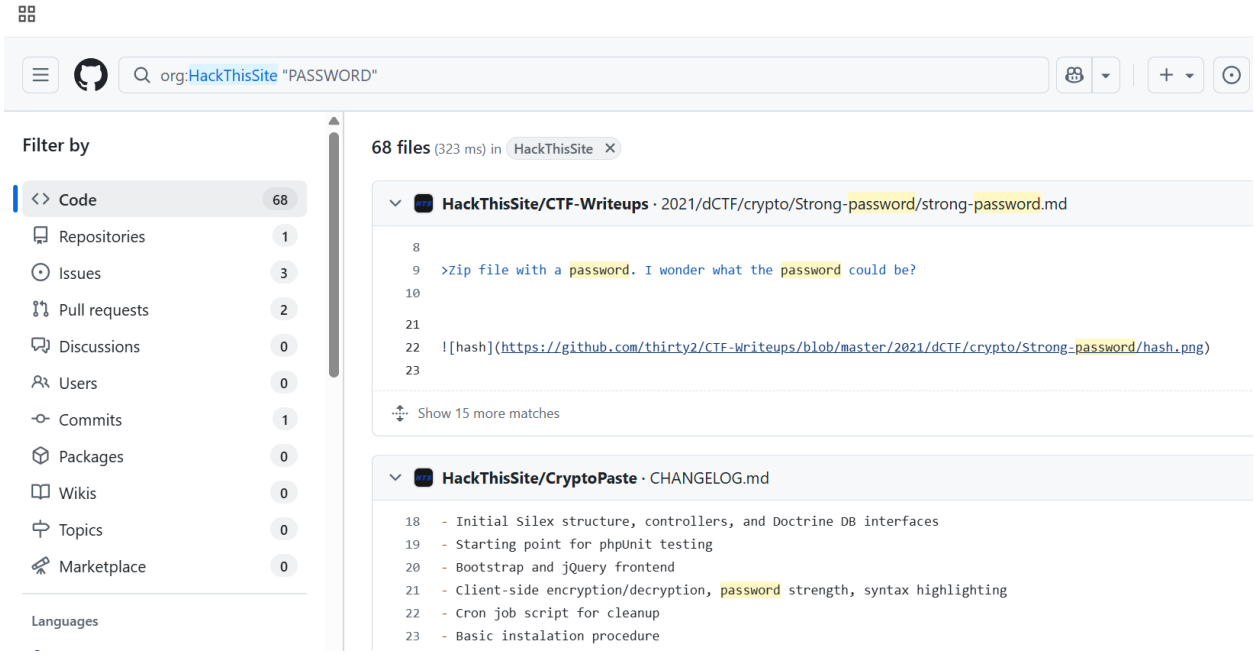


Searching for Password References:

The following query was used to identify password references:

org:HackThisSite "PASSWORD"

The search results showed multiple files containing references to passwords within documentation and code examples. These results illustrate how password-related variables may appear in repositories and why developers should avoid storing real credentials in source code.

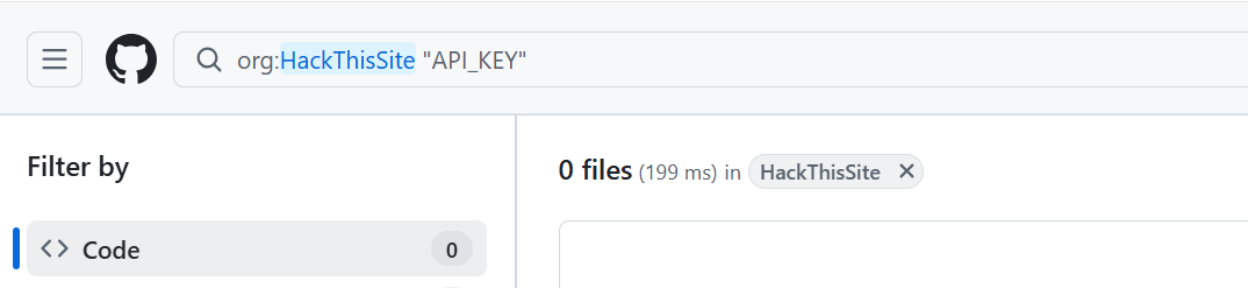


Searching for API Keys:

The following query was used to search for API keys:

org:HackThisSite "API_KEY"

No results were found for this query within the HackThisSite organization repositories. This indicates that no API keys were publicly exposed in the scanned repositories at the time of the search.



- The search queries used in this task represent only a small subset of GitHub dorks that can be used for reconnaissance.
- Many additional dorks exist that can help identify exposed credentials, API keys, configuration files, and tokens.
- Security researchers often combine multiple search queries to improve the chances of discovering sensitive information.
- GitHub dorking techniques can be adapted depending on the target organization and the type of data being investigated.
- Proper repository management and secret protection are important to prevent accidental exposure of sensitive information.

Conclusion:

The GitHub Dorking process demonstrated how sensitive information can be discovered in public repositories if secure coding practices are not followed. By using targeted search queries, it is possible to identify potential credential leaks and configuration exposures. Organizations should ensure that secrets are properly protected and avoid storing sensitive data directly in public repositories.

Task 4 – Live Asset Filtering & Visual Recon

Introduction:

During reconnaissance, many subdomains are discovered through enumeration tools. However, not all discovered subdomains are active or reachable. Some may be inactive, parked, or no longer in use. Live asset filtering helps identify which subdomains are actually responding to HTTP or HTTPS requests. This allows security researchers to focus their analysis only on active assets.

Objective:

The objective of this task is to filter out inactive subdomains and identify live web assets. After identifying the active hosts, screenshots of the web pages are captured to visually analyze the attack surface and detect interesting components such as login panels, admin dashboards, or development environments.

Tools Used:

httpx: Used to check which subdomains respond to HTTP or HTTPS & Gather additional information such as status codes and page titles

Aquatone: Used to capture screenshots of live web assets

Gowitness: Alternative tool for visual reconnaissance

Methodology:

Filter Live Subdomains:

After collecting subdomains in **Task 1**, the list was filtered to identify active hosts.

Command used:

cat allattacks.txt | httpx -silent -o live-urls.txt

```
┌──(kali㉿kali)-[~/Week 1]
└─$ cat allattacks.txt | httpx -silent -o live-urls.txt
https://hackthissite.com
https://ctf.hackthissite.org
https://qdb.hackthissite.org
https://stats.hackthissite.org
https://hacked.chat
https://hackmail.org
https://h5ai.hackthissite.org
https://irc.hackthissite.org
https://legal.hackthissite.org
http://psrp.hackthissite.org
https://mirror.hackthissite.org
https://irc-lille.hackthissite.org
https://irc-www.hackthissite.org
https://hacksite.net
https://lille.irc.hackthissite.org
https://irc-wolf.hackthissite.org
https://forums.hackthissite.org
https://status.hackthissite.org
https://hackthissite.org
https://www.hackthissite.com
https://wolf.irc.hackthissite.org
https://hp.hackthissite.org
https://www.irc.hackthissite.org
https://www.hackthissite.org
http://4-v3stage-cdn.hackthissite.org
http://4-v3dev-cdn.hackthissite.org
http://1-v3dev-cdn.hackthissite.org
http://1-v3stage-cdn.hackthissite.org
http://0-v3stage-cdn.hackthissite.org
http://irc-hub.hackthissite.org
http://2-v3dev-cdn.hackthissite.org
http://0-v3dev-cdn.hackthissite.org
http://hub.irc.hackthissite.org
http://3-v3stage-cdn.hackthissite.org
http://2-v3stage-cdn.hackthissite.org
http://3-v3dev-cdn.hackthissite.org
http://v3stage-cdn.hackthissite.org
http://v3dev.hackthissite.org
http://v3dev-cdn.hackthissite.org
```

Collect Additional Information:

Another command was used to gather more details about live assets.

cat allattacks.txt | httpx -status-code -title -tech-detect -o live-httpx.txt

```
┌──(kali㉿kali)-[~/Week 1]
└─$ cat allattacks.txt | httpx -status-code -title -tech-detect -o live-httpx.txt

    projectdiscovery.io

[INF] Current httpx version v1.8.1 (latest)
[WRN] UI Dashboard is disabled, Use -dashboard option to enable
https://lille.irc.hackthissite.org [301]
https://hackthissite.com [200]
https://mirror.hackthissite.org [200] [index - powered by h5ai v0.29.2 (https://larsjung.de/h5ai/)] [HSTS,PHP,h5ai]
https://hacksite.net [302] [HSTS]
https://ctf.hackthissite.org [200] [35C3 Junior CTF Write-ups Posted | HackThisSite CTF] [Fastly,GitHub Pages,Jekyll:3.7.4,Ruby,Varnish]
https://qdb.hackthissite.org [200] [HackThisSite IRC] [HSTS]
https://hackthissite.org [200] [Hack This Site] [HSTS,jQuery:1.8.1]
https://irc-lille.hackthissite.org [301]
https://legal.hackthissite.org [200] [HackThisSite Legal] [HSTS,Modernizr,jQuery]
https://hacked.chat [302] [HSTS]
https://irc-www.hackthissite.org [200] [HackThisSite IRC] [HSTS]
https://stats.hackthissite.org [200] [Sign in -                    Piwik] [AngularJS,HSTS,Matomo Analytics,jQuery]
https://hackmail.org [302]
https://irc.hackthissite.org [301]
https://irc-wolf.hackthissite.org [301] [301 Moved Permanently] [HSTS,Nginx:1.18.0,Ubuntu]
https://h5ai.hackthissite.org [200] [HackThisSite / h5ai] [Fastly,GitHub Pages,Varnish]
https://status.hackthissite.org [302] [HSTS]
http://psrp.hackthissite.org [404] [404 - redirect.pizza]
https://hp.hackthissite.org [200] [Hack This Site] [HSTS,jQuery:1.8.1]
https://www.hackthissite.com [200]
https://wolf.irc.hackthissite.org [301] [301 Moved Permanently] [HSTS,Nginx:1.18.0,Ubuntu]
https://www.irc.hackthissite.org [200] [HackThisSite IRC] [HSTS]
https://www.hackthissite.org [200] [Hack This Site] [HSTS,jQuery:1.8.1]
https://forums.hackthissite.org [403] [403 Forbidden] [HSTS]
http://1-v3stage-cdn.hackthissite.org [400] [ERROR: The requested URL could not be retrieved]
http://0-v3stage-cdn.hackthissite.org [400] [ERROR: The requested URL could not be retrieved]
http://0-v3dev-cdn.hackthissite.org [400] [ERROR: The requested URL could not be retrieved]
http://2-v3stage-cdn.hackthissite.org [400] [ERROR: The requested URL could not be retrieved]
```

Capture Screenshots of Live Assets:

After identifying live URLs, screenshots were captured using Aquatone.

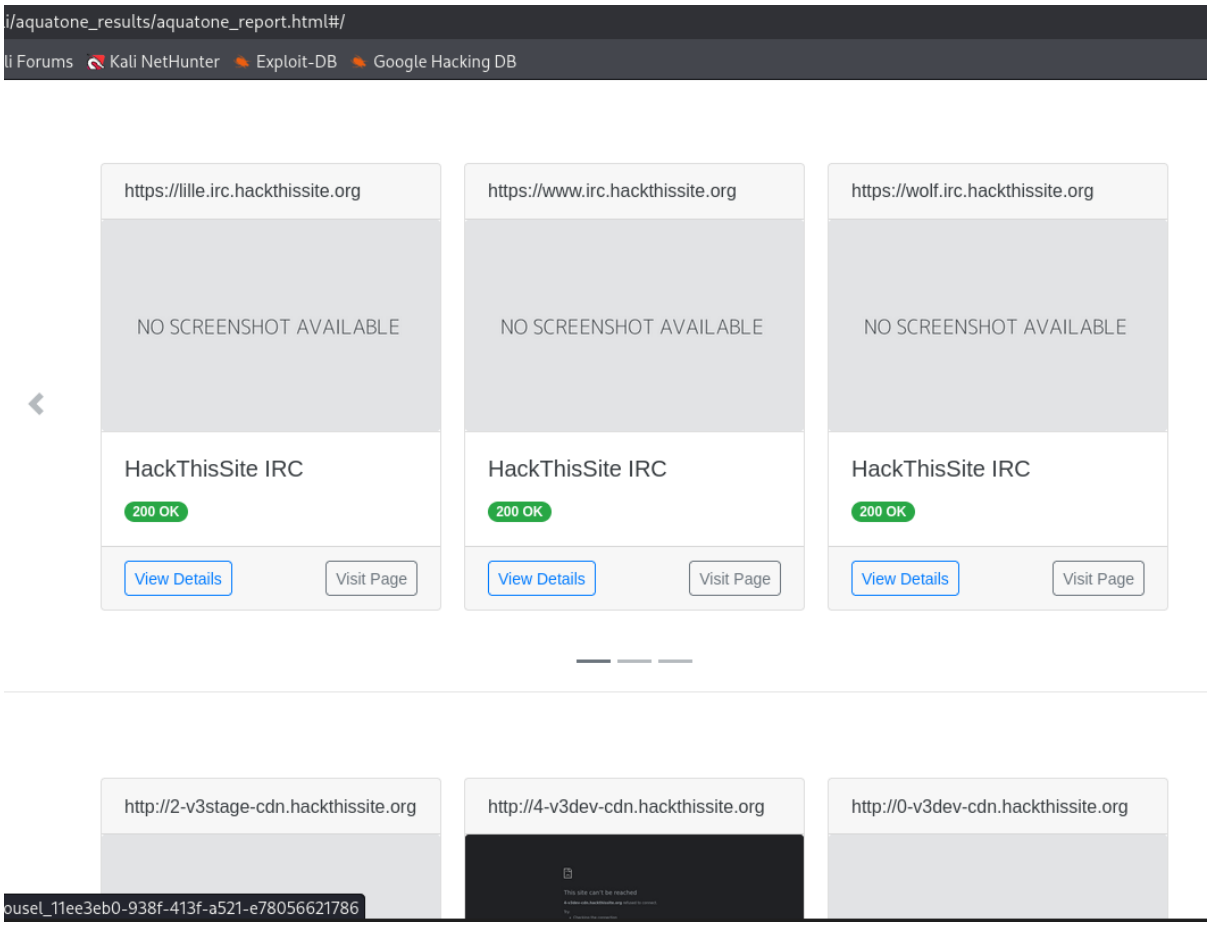
cat live-urls.txt | aquatone -out ~/aquatone_results

```
┌──(kali㉿kali)-[~/Week 1]
└─$ cat live-urls.txt | aquatone -out ~/aquatone_result
aquatone v1.7.0 started at 2026-03-07T11:42:54-05:00

Targets    : 40
Threads    : 2
Ports      : 80, 443, 8000, 8080, 8443
Output dir : /home/kali/aquatone_results

https://hackthissite.com: request timeout
http://4-v3dev-cdn.hackthissite.org: 400 Bad Request
https://hackmail.org: request timeout
https://irc-www.hackthissite.org: 200 OK
https://hacksite.net: 200 OK
https://irc-wolf.hackthissite.org: 200 OK
https://hacked.chat: 200 OK
http://1-v3dev-cdn.hackthissite.org: 400 Bad Request
https://irc.hackthissite.org: 200 OK
https://hp.hackthissite.org: request timeout
http://0-v3stage-cdn.hackthissite.org: 400 Bad Request
http://irc-hub.hackthissite.org: request timeout
http://0-v3dev-cdn.hackthissite.org: 400 Bad Request
https://mirror.hackthissite.org: 200 OK
http://2-v3stage-cdn.hackthissite.org: 400 Bad Request
https://legal.hackthissite.org: 200 OK
https://ctf.hackthissite.org: 200 OK
```

```
https://irc-wolf.hackthissite.org: screenshot successful
Calculating page structures... done
Clustering similar pages... done
Generating HTML report... done

Writing session file...Time:
 - Started at  : 2026-03-07T11:42:54-05:00
 - Finished at : 2026-03-07T11:44:24-05:00
 - Duration    : 1m30s

Requests:
 - Successful : 33
 - Failed     : 7

 - 2xx : 17
 - 3xx : 0
 - 4xx : 16
 - 5xx : 0

Screenshots:
 - Successful : 2
 - Failed     : 31

Wrote HTML report to: /home/kali/aquatone_results/aquatone_report.html
```

- The generated report can be opened using:

firefox ~/aquatone_results/aquatone_report.html



Findings & Analysis:

- Several subdomains were tested to identify which hosts were actively responding to HTTP or HTTPS requests.
- The httpx tool helped identify live URLs along with status codes and page titles.
- Screenshot tools such as Aquatone allowed visual inspection of web interfaces.
- Some discovered assets included login pages and application interfaces which may require further security testing.

Conclusion:

Live asset filtering and visual reconnaissance are important steps in the reconnaissance process. By identifying active hosts and capturing screenshots of web interfaces, security researchers can efficiently analyze the exposed attack surface of a target. This approach saves time and helps prioritize the most relevant assets for further security analysis.